GAUSS LENS: Auditable 3D Face Parameters from a Single Photo

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Abstract

Neural face trackers return numbers such as head pose or mouth state, but they cannot say why those numbers should be trusted, and they cannot refuse a bad photo in a way that can be inspected. GAUSS LENS splits the job. A neural model (Meta SAM 3.1) does the perception: it finds the face, eyes, eyebrows, nose and lips. A Gauss-Newton solver written in plain NumPy, with no neural network inside it, fits a 3D morphable face model to those regions and grades its own result with five quality scores and a confidence value. A second neural model (Meta Muse Spark) writes a short verdict from the solver's report, and the verdict is discarded if it quotes a number the solver did not produce. On a frontal portrait the system reaches quality 0.962 (PROCEED) with a mean region error of 4.3 px, and the solver residual falls by 90.5% in ten iterations. A 79 px face and a head turned by about 40° are refused, with the failing score named.

Contributions

1. A split between neural perception and network-free geometry, enforced by a test that fails if any solver module imports an ML or API library.
2. A mask-to-landmark construction that handles merged left/right masks and reads mouth and eye openness directly from the masks, where the fitted values are poorly determined.
3. A five-score acceptance gate in which every score is a closed-form expression of the solver's own output.
4. An audit layer in which the language model can only restate numbers present in the solver's report.

1. Motivation and claim

A face tracker is only useful if its output can be checked. We therefore make a deliberately narrow claim: there is no neural network in the 3D solve. SAM and Muse Spark are neural; the step that turns landmarks into head pose and face parameters is NumPy code [20] that can be read, logged and tested. A test (`tests/test_no_network_in_solver.py`) fails if any solver module imports an ML or API library. Each run saves the residual after every iteration, five quality scores and the error per face region, so a reviewer can see why a result was accepted or refused.

1.1 Related work

Morphable models. A 3D morphable model writes a face as a linear combination of shape and expression bases [1]. Widely used instances are the Basel Face Model [2] and FLAME [3]. Our fit uses the same linear form, on a synthetic template (Section 3.2).

Learned regression. Many current systems train a network to output morphable-model coefficients or mesh vertices directly from an image [4–7]. In those systems the step from pixels to parameters is itself a network, which is what makes the output hard to audit. GAUSS LENS makes no accuracy comparison with them (Section 5).

Segmentation as perception. We use promptable segmentation only to find regions: SAM [8] and its text-prompted successor SAM 3 [9] (we run release 3.1 and cite the SAM 3 paper). The 3D geometry is solved separately. The verdict text is written by Muse Spark [10].

Inspectable models. The case for models whose reasoning can be inspected, rather than after-the-fact explanations of opaque ones, is made in [11, 12]. Our claim is narrower: only the 3D solve and the gate are network-free.

2. System overview

```
photo → SAM 3.1 (face, eye, eyebrow, nose, lips) → masks → landmarks
      → Phase 1  S2 Gauss-Newton fit | S1 validation | S3 UV texture bake | S4 identity fit
      → Phase 2  frame fit + quality gate → PROCEED / CONSERVATIVE / FALLBACK
      → Phase 3  motion-state update (built for video; trivial for one image)
      → report.json → Muse Spark verdict (numbers checked against the report)
```

The order inside Phase 1 matters: the fit (S2) runs first, because validation (S1) needs the fitted landmarks and the fitted rotation. The system ships as a Streamlit app and a command-line tool. A new image costs five live SAM calls and one live Muse call, about $0.014 in total, and every answer is cached on disk.

3. Method

3.1 From masks to landmarks

SAM [8, 9] is queried once per phrase, five calls per new image. Paired organs (eyes, brows) that come back as one merged mask are split into connected components, ignoring components smaller than 15% of the largest. The template is then calibrated to the face (a piecewise-linear vertical warp anchored at brow, eye, nose and mouth rows and the chin, plus a horizontal scale from the inter-ocular ratio). Each region keeps the template's own internal geometry and is translated so its centre sits on the centre of its measured mask; the chin landmark sits at the bottom centre of the face box. Ears are not used, because hair often hides them. The initial translation comes in closed form from the landmark spread, $Z = f \cdot \sigma_{model} / \sigma_{image}$.

Concretely, let c_r be the template centre of region r, (u_r, v_r) the centre of its measured box, and $s = f / Z$ the pixel scale. The landmark of vertex i in region r is:

$$x_i = (u_r + s \cdot (S_{ix} - c_{rx}) \ , \ v_r - s \cdot (S_{iy} - c_{ry})), \ i \in V_r$$

On the built-in model this gives 27 landmarks: 8 eye, 4 brow, 5 nose, 9 mouth and one chin. Two quantities are measured directly from the masks, so they cannot run away in a fit: eye aspect (mask height ÷ width) and mouth aspect (lips-mask height ÷ width, open if above 0.5).

3.2 The fit (Stage 2)

The face is a 3D morphable model [1] with regional sub-models (S_0 is the mean shape):

$$S = S_0 + A_{id} \cdot \alpha + A_{exp} \cdot \beta$$
$$S_{eye} = S_{eye_ref} + B_{eye} \cdot \psi_{eye} \quad S_{brow} = S_{0,brow} + B_{brow} \cdot \gamma_{brow}$$
$$S_{nose} = S_{0,nose} + \delta_{nose} \quad S_{mouth}, \ S_{ear}: \text{ same form with } \psi_{mouth}, \ \gamma_{ear}$$
$$\pi(S_i; \theta) = (1/Z_i) \cdot K \cdot (R(\omega) \cdot S_i + t), \ Z_i = [R(\omega) \cdot S_i + t]_z \ (R = \text{Rodrigues})$$

The projection π is the standard pinhole model [16]. The built-in template is a 25 × 25 grid mesh (625 vertices, 1,152 triangles) with a generated basis, not a scanned statistical model. Only the first 20 identity and the first 15 expression coefficients are active in the fit, and the ear parameters stay at zero because ears are not requested from SAM. The solver minimises a three-term energy over $\Theta = [\alpha \ \beta \ \omega \ t \ \psi_{eye} \ \gamma_{brow} \ \delta_{nose} \ \psi_{mouth} \ \gamma_{ear}]$:

$$E(\theta) = w_{app} \cdot E_{app} + \Sigma_i \ \| \pi(S_i; \theta) - x_i \|^2 + E_{reg}$$
$$E_{app} = \Sigma_i \ \| I(\pi(S_i)) - T_i \|^2, \quad E_{reg} = \| L_{tps} \cdot \delta_{nose} \|^2$$
$$\Delta\theta = -(J^T J + \lambda \Lambda)^{-1} \ J^T \ e \ , \quad \lambda = 0.01$$

Here x_i is the landmark from the SAM masks, T_i is the image colour sampled at that landmark, $w_{app} = 5$, and L_{tps} is the graph Laplacian of the nose sub-mesh (a smoothness prior). Λ is a block-diagonal damping prior (Levenberg–Marquardt-style damping [14, 15]; see [13] for Gauss-Newton methods) with a separate strength for each block (identity and expression, pose, eye, brow, mouth corners, ear, nose smoothness plus a nose anchor, and a depth anchor on t_z); the identity and expression blocks scale with the number of landmarks so they stay proportionally strong on denser meshes. The

Jacobian is built analytically, block by block, using the exact derivative of Rodrigues' formula [17] rather than a first-order approximation.

Each step is checked by backtracking: the step is halved up to five times until no vertex has moved behind the camera and the residual has not grown. If no safe step exists, the solver holds its position. The solver stops after 10 iterations or when an accepted step is shorter than 1e-4. Eye, brow, mouth and ear parameters are clipped to safe ranges as a last safety net, and the damping does the real work.

3.3 Validation (Stage 1)

Five scores, each in [0, 1]. Sharpness is a Laplacian-based focus measure [18]. The photo is accepted only if all five are above 0.60:

```
resolution = min(1, min(H_face, W_face) / 256)
sharpness = min(1, ‖∇²I‖ / ‖∇²I_ref‖) (8-neighbour Laplacian vs a synthetic sharp reference)
pose = exp(−(ω_y² + ω_x²) / 2σ²), σ = 30°
symmetry = 1 − ‖L_left − flip(L_right)‖ / (k·W_face), k = number of left/right landmark pairs
landmark = (1/N) Σ_i exp(−‖x_i − x̂_i‖² / 2σ_lm²), σ_lm = max(3 px, 0.02·W_face)
```

Because 0.60 × 256 ≈ 154, a face must be at least about 154 px across. Pose and landmark scores use the fitted ω and the fitted landmarks $\hat{x}_i$, which is why the fit comes first.

3.4 Quality gate (Phase 2)

```
c_t = 0.5·s_lm + 0.3·s_flow + 0.2·s_pose
PROCEED if c_t > 0.75; CONSERVATIVE if 0.40 ≤ c_t ≤ 0.75; FALLBACK if c_t < 0.40
```

In the gate, s_{lm} uses $\sigma_{lm} = 12.8$ px, calibrated on the median error measured on a real run, and $s_{flow} = \exp(-\text{mean}\|f_t - f_{pred}\|^2 / 2\sigma_{flow}^2)$ with $\sigma_{flow} = 2$ px. For a single photo there is no previous frame, so $s_{flow} = 1$. Phase 2 also provides a Newtonian pose warm-start and a per-frame joint fit with the identity α frozen; Phase 3 adds Gauss-Newton optical flow [19] and a velocity update, which matter for video.

3.5 The audited verdict

The report contains only measured or solved values: validation scores, decision and quality, head pose, region errors, mask measurements, missing regions and the residual curve. Muse Spark [10] sees only this report (text only, never the image, temperature 0) and writes at most three short sentences. A reply is accepted only if every number in it matches a value in the report at the precision quoted (small integers 0–3 and the published thresholds are allowed). Otherwise it is discarded. Answers are cached by report hash and live calls are capped per process. Formally, let N(r) be every number in the report plus the published thresholds (0.60, 0.75 and 0.40). A reply is accepted only if every number x it quotes, with d decimal places, passes:

$$\forall\, x \in \text{numbers(reply)},\ x \notin \{0,1,2,3\}:\ \exists\, a \in N(r) \text{ with } |x-a| \le \tfrac{1}{2}\cdot 10^{-d}$$

In the command-line tool, a reply that fails the check, comes back empty or errors is replaced by a deterministic template verdict, and the source of the final text (model or template) is shown with it. In the app, the failure is shown as an error and no text is substituted. The language model never receives the photograph, and models whose name ends in a contributor suffix are refused, because that tier may train on what is sent.

4. Results

One frontal portrait, built-in synthetic model, SAM 3.1 landmarks and the Muse Spark verdict. Section 4.1 shows the full output of this run (Figures 1–5, Tables 4–5); the headline numbers are:

Item	Result
Decision / quality	PROCEED, 0.962

Item	Result
Head pose	yaw −0.2° · pitch +0.4° · roll −1.1°
Mouth (measured from the mask)	closed, aspect 0.225
Region error (px)	all 4.3 · eye 1.5 · brow 3.3 · nose 3.9 · mouth 5.8
Solver residual	118.2 → 11.3 (90.5% lower), most of the drop in the first three iterations

Table 1. Headline result for the frontal portrait.

All five gate scores on each input (lm = landmark consistency, sym = symmetry, res = resolution, blur = sharpness); the lowest score decides:

Input	lm	sym	res	blur	pose	Decision
Sharp frontal face	0.982	0.998	1.000	1.000	1.000	Accepted, quality 0.962
Same face, blur σ 8	0.956	0.994	1.000	0.681	1.000	Accepted, quality 0.928
Same face, blur σ 14	0.975	0.994	1.000	0.627	1.000	Accepted, quality 0.965
Face about 79 px wide	0.878	0.989	0.309	1.000	1.000	Rejected on resolution
Head turned about 40°	0.336	0.894	1.000	1.000	0.994	Rejected on landmark consistency

Table 2. Gate scores for the five test inputs.

Splitting merged left/right brow masks (SAM first returned both eyebrows as one mask, which put the brow landmarks on the bridge of the nose) cut the brow error from 10.4 px to 3.3 px and the overall error from 5.8 px to 4.3 px. Landmark consistency rose from 0.843 to 0.982, symmetry from 0.929 to 0.998 and quality from 0.937 to 0.962.

Convergence. The residual falls from 118.2 to 31.2 after the first step, 16.7 after the third and 11.3 after the tenth. The step norm falls from 7.58 to 0.52 but does not reach the 1e-4 stopping threshold, so the fit stops at the iteration cap, and the report says so.

Forcing a rejected photo through. Running the 40° photo past its rejection (Strict off) gives CONSERVATIVE with quality 0.572, a mean landmark error of 47.1 px (eye 99.8 px) and 21 landmarks, because one eye was not found. The reported yaw is only −6.3°, and the verdict states that validation failed and the parameters are unreliable.

These are demonstrations of behaviour on a handful of photos, not benchmark statistics. The faces are synthetic, except one real photograph with a small face, used only locally for the resolution check and not distributed.

4.1 Worked example: the full output for one photo

The app and command-line tool write every intermediate value of a run, so the numbers in Table 1 can be traced back to files. This section shows the output of the single frontal-portrait run (synthetic face, built-in model). The same values are stored in `report.json` and `summary.json`; nothing below is computed for the figure.



Figure 1. Fitted landmarks drawn on the input portrait (one colour per region: brow, eye, nose, mouth; the four purple markers are the template ear points, which are not requested from SAM and are not used in the fit). Each region keeps the template's own shape and is centred on its mask (Section 3.1), so the markers show where the model places each region, not traced feature outlines. This is why a 4.3 px mean error shows agreement with the SAM-derived input and is not an accuracy claim (Section 5).

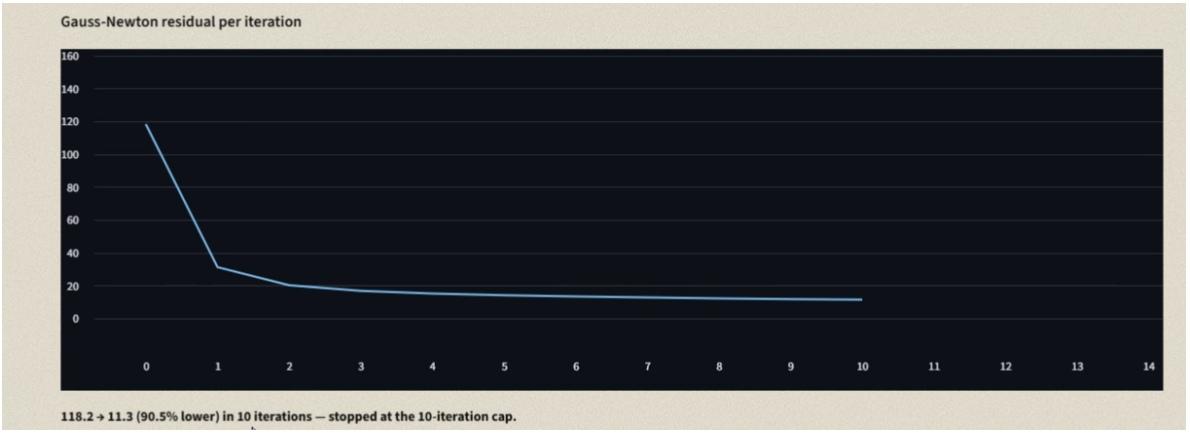


Figure 2. Gauss-Newton residual per iteration. The residual falls from 118.2 to 11.3 (90.5% lower) in 10 iterations and the fit stops at the 10-iteration cap, not at convergence. Values per iteration are in Table 3.

Iteration	Residual before	Residual after	Step norm	Step accepted
0	118.208	31.184	7.581	yes
1	32.137	20.053	2.432	yes
2	21.029	16.747	1.373	yes
3	17.424	15.104	0.973	yes
4	15.531	14.063	0.774	yes
5	14.321	13.295	0.657	yes
6	13.551	12.667	0.597	yes
7	13.078	12.133	0.561	yes
8	12.774	11.669	0.537	yes
9	12.575	11.262	0.521	yes

Table 3. Residual history from `report.json`. The residual before an iteration is slightly higher than the residual after the previous one in every row; the two are not identical numbers, and the step norm is still 0.52 at the end.



Figure 3. Head pose and mean landmark error per region (px, lower is better): all 4.3, brow 3.3, eye 1.5, mouth 5.8, nose 3.9. The pose of −0.2° / +0.4° / −1.1° is plausible for a frontal portrait, but on turned faces the fitted yaw must not be read as a measurement (Section 5).

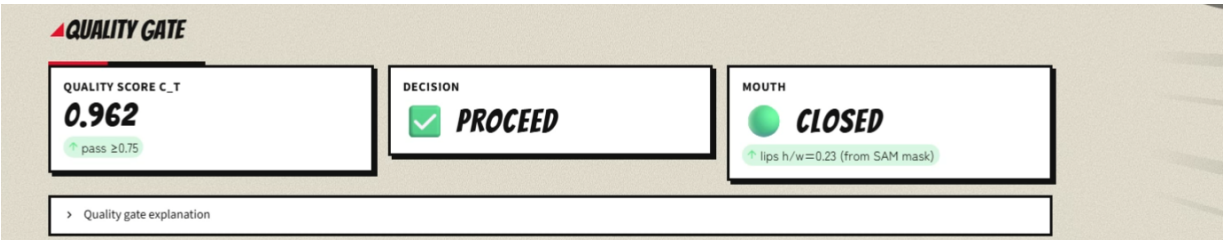


Figure 4. Quality gate: c_t = 0.962 is above the 0.75 threshold, so the decision is PROCEED. The mouth state (closed, lips height/width 0.23) comes from the SAM lips mask, not from the fitted mouth parameters.

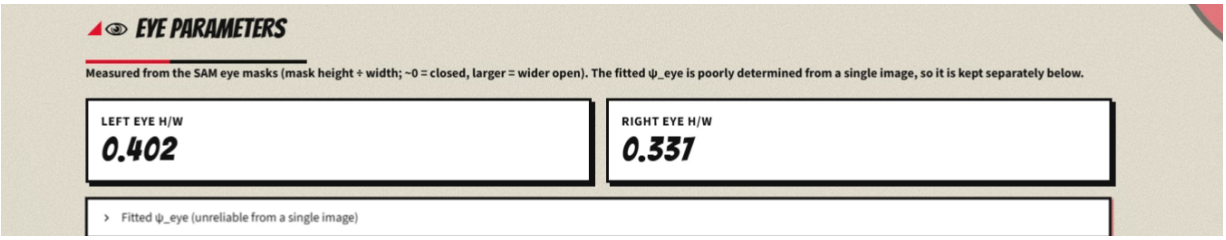


Figure 5. Eye openness measured from the SAM eye masks (height ÷ width: left 0.402, right 0.337). The fitted ψ_{eye} is kept separately and marked unreliable, because one image does not determine it (the fitted aperture is 0 on both open eyes).

Quantity	Value	Source
Eye aspect, left / right	0.402 / 0.337	SAM eye masks
Mouth aspect (lips height ÷ width)	0.225 (closed; open if above 0.5)	SAM lips mask
Regions missing	none	report.json
Weakest validation score	landmark consistency, 0.982	report.json
SAM cost	5 live calls, $0.0125	summary.json
Muse Spark cost	1 live call, $0.00174	summary.json
Total cost of the run	$0.01424	summary.json

Table 4. Measured values and cost of the run.

The saved verdict for this run (`verdict.txt`) is three sentences, and every number in it (0.982, −0.2, 0.4, −1.1, 0.225, 10 of 10) appears in the report:

> *Decision is PROCEED, with the weakest validation check being landmark_consistency at 0.982. Head pose is yaw −0.2, pitch 0.4, roll −1.1 deg, and the mouth is closed with mouth_aspect 0.225. The solver stopped at its iteration cap at 10 of 10 iterations.*

Table 5. Verdict text as saved. The audit check of Section 3.5 passes for each number it quotes.

The remaining app outputs (expression delta, mouth and nose parameters) are shown in Appendix A for completeness. They are not part of any accuracy claim.

5. Limitations and future work

- **The face model is synthetic.** All fits use a template mesh with a generated basis, not a scanned statistical model. The claim is an auditable fit with measured residuals, not an accurate 3D reconstruction of a real face. Models in BFM or FLAME form [2, 3] can be loaded as .npz.
- **Low landmark error is partly built in.** Each region keeps the template's shape and is centred on its mask, so a small error shows the solver agrees with its own input, not that the 3D shape is independently correct. A fair test needs ground-truth 3D data; more points per region from the mask outline would also help.
- **SAM is part of the causal chain.** It selects the landmark positions and, through calibration, the template proportions. What is network-free is the solver and the gate, not the whole result.
- **Head turn is detected by rejection, not measured.** On the 40° photo the pose score stayed at 0.994 and the forced fit reported a yaw of only −6.3°; the photo was rejected because the landmark and symmetry scores fell. The fitted yaw must not be shown as a measurement on turned faces.
- **Part of the parameter space is not determined by one image.** In the per-frame fit the expression block had numerical rank 38 of 64 (condition number above 10^{32}), the fitted mouth value saturated at its bound, and the fitted eye aperture read zero on open eyes. This is why eye and mouth state are measured from the masks and the fitted values are kept as unreliable in the JSON.
- **Not fully converged.** The residual is still falling at iteration ten (step norm 0.52); the report and verdict say so. The source fit is also repeated several times in the current code, which costs time but not accuracy.
- **Lenient sharpness gate.** Sharpness is compared with a synthetic reference, so a blur of σ 14 still scored 0.627 against 0.60, and a naturally soft photo can score low. The gate's σ_{lm} of 12.8 px is a snapshot of current accuracy, not a target.
- **Mask confidence is unused.** SAM's mask score did not come through the parser (it read zero), so all landmarks count equally. Once it does, it becomes the weight w_i in the geometric term.
- **Small sample.** The experiments are demonstrations on a few photos, and the baked UV texture has artefacts and is not part of any claim.

Next steps, in order of expected value: use SAM mask scores as landmark weights; replace the template with a scanned statistical model and add an independent 3D ground-truth check; measure head turn from the landmark geometry; remove the repeated source fit and raise the iteration cap; calibrate the sharpness threshold on real blur; extend to video with the per-frame tracker.

6. Responsible use

GAUSS LENS is a research prototype. It is not intended for identifying people, tracking individuals or surveillance. Use synthetic faces or faces of people who have given consent, and keep photos, `.env` and `cache/` out of version control.

Symbols

Symbol	Meaning
$\alpha \in R^{80}$, $\beta \in R^{64}$	identity and expression coefficients
$\omega \in R^3$, $t \in R^3$	rotation vector (Rodrigues) and translation
$\psi_{eye} \in R^6$	per-eye aperture and horizontal / vertical gaze
$\psi_{mouth} \in R^4$	mouth sub-model coefficients
$\gamma_{brow} \in R^8$, $\gamma_{ear} \in R^4$	brow and ear sub-model coefficients
$\delta_{nose} \in R^{3\|V_nose\|}$	per-vertex nose displacement
E_{app}, w_{app}	appearance term and its weight (5)
E_{reg}	smoothness prior on the nose, $\|L_{tps}\ \delta_{nose}\|^2$
Λ	block-diagonal damping prior
k	number of left/right landmark pairs
x_i, $\hat{x}_i$	detected and model-projected landmark

References

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Appendix A. Additional app outputs for the same photo

These panels come from the same run as Section 4.1. They show fitted parameters that one image does not determine well (Section 5), so they are given for transparency and are not used in the accept/refuse decision.

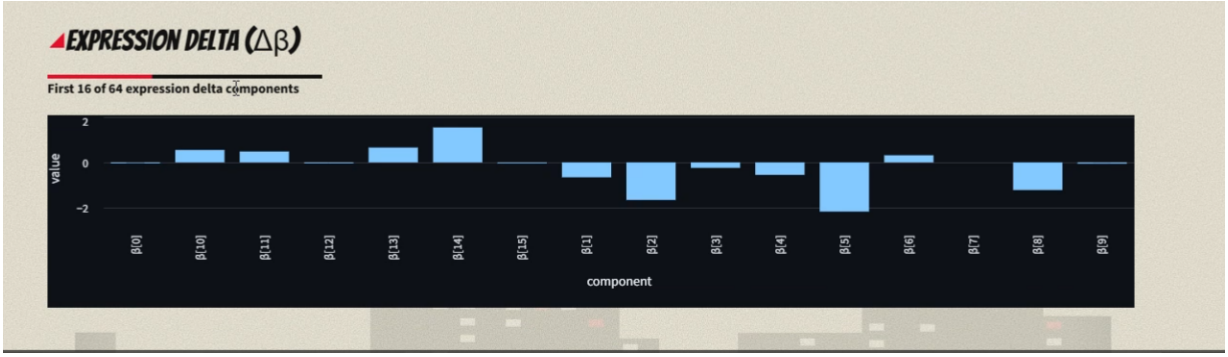


Figure A1. First 16 of the 64 expression delta components (Δβ). Only the first 15 expression coefficients are active in the fit.

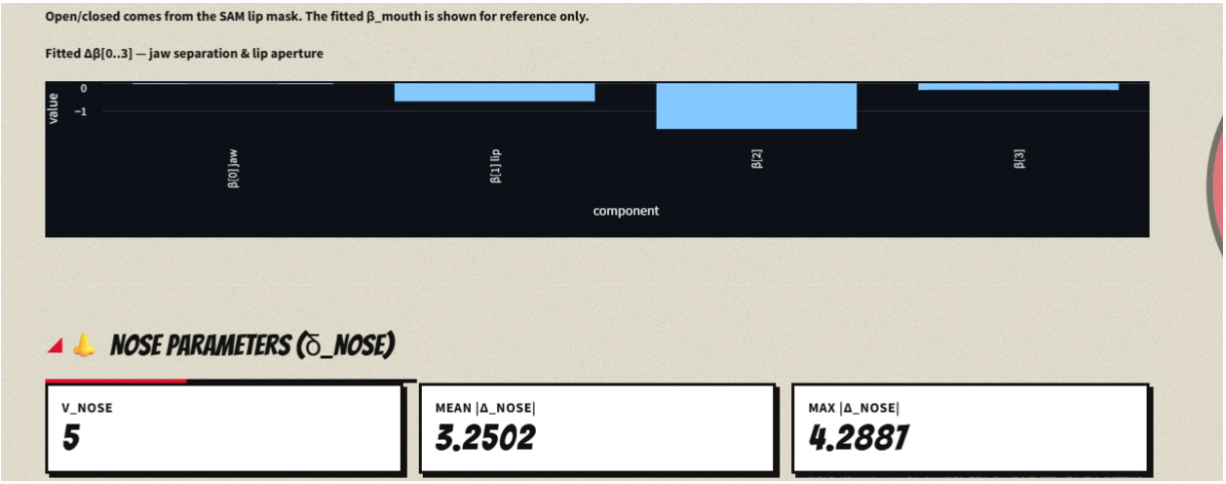


Figure A2. Fitted mouth deltas Δβ[0..3] (jaw separation and lip aperture) and the nose displacement summary: 5 nose vertices, mean $|\Delta_{nose}|$ 3.2502, max $|\Delta_{nose}|$ 4.2887. Mouth open/closed comes from the SAM lips mask, not from these fitted values.